ZZ-S-751a
NOVEMBER 5, 1959
SUPERSEDING
Fed. Spec. ZZ–S–751
November 15, 1932

FEDERAL SPECIFICATION

STOPPERS, BOTTLE, RUBBER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers rubber bottle stoppers intended for use in the chemical laboratory.

1.2 Classification.

1.2.1 *Types and grades.* — Rubber bottle stoppers shall be of the following types and grades, as specified (see 6.2):

Types:

A.—Solid.
B.—Perforated.

Grades:

I.—Laboratory.
II.—Common.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP–B–566 — Boxes, Folding, Paperboard.
PPP–B–585—Boxes; Wood, Wirebound.
PPP–B–601—Boxes, Wood, Cleated-Plywood.
PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636—Boxes, Fiber.
PPP–B–676 — Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 601—Rubber: Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–B–10377 — Box, Wood, Cleated, Veneer, Paper Overlaid.
MIL–L–10547 — Liners, Case, Waterproof.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:26Z
Check the source to verify that this is the current version before use.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Grade I.

3.1.1 *Material.*—Grade I stoppers shall be made of properly vulcanized natural rubber or synthetic rubber, or a mixture of natural and synthetic rubber. No component shall be present in such proportion as to bloom or exude from the stoppers, or as to be readily extracted by water, acid or alkaline solutions, or alcohol, under conditions of normal use.

3.1.2 *Acetone extract.*—The total acetone extract shall not exceed 5 percent of the weight of the rubber when tested as specified in 4.4.3.

3.1.3 *Hardness.*—A solid rubber stopper when tested as specified in 4.4.1 shall show an indentation of 36 to 54.

3.1.4 *Steaming test.*—When a solid stopper is subjected to the steaming test specified in 4.4.2, the diameter and length shall not change more than 10 percent.

3.2 Grade II.

3.2.1 *Material.*—Grade II stoppers shall be made of properly vulcanized natural rubber or synthetic rubber, or a mixture of natural and synthetic rubber.

3.2.2 *Hardness.* — A solid rubber stopper when tested as specified in 4.4.1 shall show an indentation of not more than 55.

3.3 **Dimensions.**—The dimensions of stopper shall be as shown in table I. A tolerance of ± 1 millimeter (mm.) for diameter at top and bottom and length of stopper, and tolerance of ± 0.5 mm. for diameter of perforations will be permitted (see 6.2).

TABLE I.—*Dimensions of stoppers*

Stopper number	Diameter of top	Diameter of bottom	Length	Diameter of perforations
	mm.	*mm.*	*mm.*	*mm.*
00	15	10	25	2
0	17	13	25	3
1	19	14	25	4
2	20	16	25	5
3	24	18	25	5
4	26	20	25	5
5	27	23	25	5
6	32	26	25	5
6½	34	27	25	5
7	37	30	25	5
8	41	33	25	5
9	45	37	25	5
10	50	42	25	5
11	56	48	25	5
12	64	54	25	5
13	68	58	25	5
14	90	75	39	5
15	103	83	39	5
Extra long size				
20	22	17	26	5
21	25	19	32	5
22	26	20	32	5
23	28	22	32	5
24	30	23	33	5
25	34	28	38	5
26	40	34	38	5
Kjeldahl stoppers				
500 ml	32	26	34	8
800 ml	39	28	35	8

3.4 **Perforations.**—When specified, stoppers shall be furnished with one, two, or three perforations. These perforations shall be parallel to the axis of the stopper and symmetrically situated with reference to it. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:26Z
Check the source to verify that this is the current version before use.

diameter of perforations shall be as shown in table I.

3.5 Identification marking.—Each stopper shall bear an identification of its size in raised numbers.

3.6 Workmanship.—Workmanship shall be first class throughout. Stoppers shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection.—Such examinations and tests as are set forth in this specification or as shall otherwise be appropriate or necessary to insure that supplies conform to specification requirements, shall be performed by and at the expense of the supplier. Suppliers who do not have facilities adequate for such tests shall arrange for the use of test facilities acceptable to the Government. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request, at any time, or from time to time, for a period of 2 years after delivery of the supplies to which such records relate.

4.1.1 Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc. investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.*—When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 *For visual examination.*—Sampling for visual examination shall be conducted in accordance with Military Standard MIL–STD–105, using inspection level I. The sample unit shall be one stopper.

4.2.2 *For tests.*—Sampling for tests shall be conducted in accordance with Military Standard MIL–STD–105, using inspection level L4, except that a minimum of 10 samples for all lot sizes shall be taken. The sample unit shall be one stopper.

4.3 Visual examination.—The samples selected in accordance with 4.2.1 shall be inspected for dimensions, perforation, marking, and workmanship requirements as specified in section 3. The AQL shall be 1.5 defects per 100 units. Any departure from the perforation, marking, workmanship, and dimensional requirements shall be considered a defect.

4.4 Tests.—The samples selected in accordance with 4.2.2 shall be tested for hardness (see 3.1.3 and 3.2.2), steaming (see 3.1.4), and acetone extract (see 3.1.2) as specified in 4.4.1, 4.4.2, and 4.4.3, respectively. Where possible, the same sample unit shall be used for the determination of two or more test characteristics. The acceptance number shall be zero (0). Any departure from the requirements specified in 3.1.3, 3.2.2, 3.1.4, and 3.1.2 shall be considered a defect.

4.4.1 *Hardness.*—Stopper shall be tested for hardness in accordance with method 3021 of Federal Test Method Standard No. 601.

4.4.2 *Steaming.*—The steaming test shall be carried out in an autoclave or other suitable containers, the temperature and pressure of which shall be controlled. A solid stopper is suspended in an atmosphere of steam at 100° centigrade for 5 days, then taken out and kept at room temperature for 24 hours.

4.4.3 *Extract, acetone.*—Stoppers shall be tested for acetone-soluble material in accordance with method 16321 of Federal Test Method Standard No. 601.

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of the levels of pack-

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:26Z
Check the source to verify that this is the current version before use.

aging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Packaging quantities.* — The number of units indicated in table II shall be packaged in each unit, intermediate, and exterior container, as applicable for the required level of protection specified in military procurement documents.

5.1.1.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.2 *Levels A and B (for military procurement).*

5.1.2.1 *Intermediate package.*—The number of units (stoppers), specified in column 3 of table II, shall be packaged in a kraft paper or cloth bag, or a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566, PPP–B–636, or PPP–B–676. Each carton shall be adequately secured to prevent accidental opening.

5.1.3 *Level B (for civil agency procurement).*

5.1.3.1 *Unit package.*—One (1) pound of stoppers of the same type and grade shall be packaged in a paper sack; or in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Each carton or sack shall be adequately secured to prevent accidental opening.

5.1.3.2 *Intermediate packaging.*—Five (5) pounds of stoppers of the same type and grade, packaged as specified in 5.1.3.1, shall be intermediately packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566, PPP–B–676, or PPP–B–636.

5.1.4 *Level C.*—Rubber stoppers shall be packaged in standard commercial containers of the size and kind commonly used, which

TABLE II.—*Packaging and packing quantities*
(Applicable to Levels A and B for military procurement)

Type	Stopper No.	No. of units (stoppers) to be contained in each intermediate package	No. of intermediate packages to be contained in exterior container	Total number of units (stoppers) in each exterior container
A	00	100	24	2,400
A	0	100	24	2,400
A	1	100	18	1,800
A	2	100	18	1,800
A	3	100	18	1,800
A	4	100	18	1,800
A	5	100	14	1,400
A	6	50	20	1,000
A	8	50	10	500
A	10	25	16	400
B	0	100	24	2,400
B	1	100	18	1,800
B	2	100	18	1,800
B	4	100	18	1,800
B	5	100	14	1,400
B	6	50	20	1,000
B	7	50	20	1,000

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:26Z
Check the source to verify that this is the current version before use.

will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, table III; or Military Specification MIL–B–10377, table II.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specifications.

Note.—Strapping will not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B (for military procurement).* —Exterior container shall conform to Federal Specification PPP–B–601, table I, using type III, class 1 plywood, or shall be designed for type 2 load, and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 2 use; PPP–B–621, table I; PPP–B–636, type I, class 1, fabricated from not less than 275-pound test fiberboard; or Military Specification MIL–B–10377, table I.

5.2.3 *Level B (for civil agency procurement).* — Stoppers of the same type and grade, packaged as specified in 5.1.3.1 and intermediately packaged as specified in 5.1.3.2, shall be packed in fiberboard shipping containers conforming to Federal Specification PPP–B–636, type I or II, class 1, style optional. Closure and sealing shall conform to the appendix to the box specification. Weight limitation of the shipping containers shall be 50 pounds.

5.2.4 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Civil agencies.*—In addition to any special marking in the contract or order, marking shall be in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*

5.3.2.1 *Intermediate packages.*—Each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2.2 *Exterior container.*—Exterior container shall be marked in accordance with Military Standard MIL–STD–129, except that 6.2(c), 6.2(d), 6.5 and 7.5.1 shall not be applicable. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Intended use.

6.1.1 Grade I stoppers are intended for general use in the chemical laboratory. They are designed for good mechanical performance, good aging and reasonably free from components which might contaminate aqueous and alcoholic solutions.

6.1.2 Grade II stoppers are intended for use when low initial cost is important and possible contamination by sulfur or materials coming in contact with stoppers is of little consequence. Stoppers of this grade are also intended for use when ease of cutting and boring is desired.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:26Z
Check the source to verify that this is the current version before use.

6.2 **Ordering data.** — Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Number and date of this specification.
(*b*) Type, grade, and size of stoppers (see 1.2 and 3.3).
(*c*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(*d*) Marking of intermediate packages, if desired (see 5.3.1).

6.3 This specification covers the following items appearing in the Armed Services Medical Stock List:

8125–440–7500 STOPPER, BOTTLE, Rubber, One Hole, No. 0.
8125–440–8000 STOPPER, BOTTLE, Rubber, One Hole, No. 1.
8125–440–9000 STOPPER, BOTTLE, Rubber, One Hole, No. 2.
8125–441–0000 STOPPER, BOTTLE, Rubber, One Hole, No. 4.
8125–441–1000 STOPPER, BOTTLE, Rubber, One Hole, No. 6.
8125–441–2000 STOPPER, BOTTLE, Rubber, Solid, No. 00.
8125–441–3000 STOPPER, BOTTLE, Rubber, Solid, No. 0.
8125–441–4000 STOPPER, BOTTLE, Rubber, Solid, No. 1.
8125–441–5000 STOPPER, BOTTLE, Rubber, Solid, No. 2.
8125–441–5500 STOPPER, BOTTLE, Rubber, Solid, No. 3.
8125–441–6000 STOPPER, BOTTLE, Rubber, Solid, No. 4.
8125–441–6500 STOPPER, BOTTLE, Rubber, Solid, No. 5.
8125–441–7000 STOPPER, BOTTLE, Rubber, Solid, No. 6.
8125–441–9000 STOPPER, BOTTLE, Rubber, Solid, No. 8.
8125–441–9500 STOPPER, BOTTLE, Rubber, Solid, No. 10.
8125–442–0000 STOPPER, BOTTLE, Rubber, Two Hole, No. 5.
8125–442–1000 STOPPER, BOTTLE, Rubber, Two Hole, No. 6.
8125–442–2000 STOPPER, BOTTLE, Rubber, Two Hole, No. 7.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.5 ***Transportation description.***—Transportation description applicable to this item is:

Bottle Stoppers NOIBN
Carload minimum weight 30,000 pounds per Uniform Freight Classification.
Truckload minimum weight 30,000 pounds per National Motor Freight Classification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
Navy—Md
Air Force.

U.S. GOVERNMENT PRINTING OFFICE : 1982 - 361-705/1351

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:26Z
Check the source to verify that this is the current version before use.